Inventory Operations Analysis

Excel and PostgreSQL Portfolio Project

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Project Overview

This portfolio project uses Excel and PostgreSQL to analyze 159 inventory and order records from January through June 2026 across Dallas, Tulsa, and Oklahoma City. The goal was to identify stock shortages, compare supplier delivery performance, summarize inventory value, and examine monthly order activity.

I cleaned the data, built Excel PivotTables and charts, and used SQL to validate the findings. During validation, I corrected charts that included missing late-day values as zeros. The corrected delivery analysis uses 146 records with available late-day values.

This is a practice project. Its findings describe the supplied dataset and do not represent measured results for an actual company.

Key Findings

• Stock availability: 45 of 159 records (28.3%) were below their reorder points. Control Board had the most below-reorder records, with 7.

• Shortages: Copper Wire Spool had the largest total shortage at 639 units.

• Inventory value: Recorded inventory values totaled $846,643.45. Dallas accounted for $321,684.35 (38.0%), the highest warehouse total. Electronics accounted for $421,620.85 (49.8%), the highest category total.

• Supplier performance: Green Country Components had the highest average late days among identified suppliers, at 4.6 days. One record with an unknown supplier was excluded from this comparison.

• Open orders: 13 records (8.2%) were marked Open: 6 in Dallas, 6 in Tulsa, and 1 in Oklahoma City.

• Product order quantities: Control Board had the highest total quantity ordered, at 3,508 units. Received quantities totaled 3,415 units, leaving a 93-unit gap.

• Delivery timing: 58 records had 0–1 late days, representing 36.5% of all 159 records. Late-day values were missing for 13 records.

• Relationship analysis: Among 146 complete pairs, shortage quantity and late days showed virtually no linear relationship ($R^2 = 0.0003$).

• Monthly activity: June had the highest order count, with 40 records; April had the lowest, with 17.

Recommendations

• Review Copper Wire Spool shortages and Control Board records below reorder points to identify replenishment priorities.

• Investigate Green Country Components' delivery delays using individual orders, product mix, and delivery history before making supplier decisions.

• Follow up on the 13 open orders to confirm expected delivery dates and outstanding quantities.

• Preserve missing late-day values as missing rather than replacing them with zero. Continue checking Excel results against SQL.

Limitations

This practice dataset contains 159 records covering six months. It is too limited to establish annual seasonality or predict future performance reliably.

Thirteen records lack late-day values and were excluded from delivery charts and regression. One record has an unidentified supplier and was excluded from supplier comparisons.

Inventory values are sums across the supplied records. Without confirming that warehouse and product stock balances are unique, these totals should not be treated as an audited physical inventory valuation.

The relationship analysis measures linear association; it does not establish causes of shortages. Recommendations are proposed actions, and no business savings or operational improvements have been measured.

Tools and Methods

Excel
• Removed a duplicate record, labeled a missing supplier, and standardized category capitalization.
• Built PivotTables to summarize shortages, supplier delays, stock status, and inventory value.
• Created charts to compare products, suppliers, warehouses, and delivery timing.
• Used FILTER to exclude missing values from chart inputs and RSQ to check the linear relationship.

PostgreSQL and pgAdmin
• Used SELECT, WHERE, ORDER BY, and LIMIT to explore records.
• Used GROUP BY, HAVING, and aggregate functions to summarize results.
• Applied CASE expressions to classify stock against reorder points.
• Used date functions to summarize monthly order activity.
• Used a common table expression and RANK window function to analyze warehouse totals.
• Validated Excel findings with SQL, including record counts, percentages, shortages, and R^2.

Supporting Files

• Excel workbook containing the raw data, cleaned data, cleaning log, data dictionary, PivotTables, charts, and project summary.
• SQL scripts 01–12 containing analysis exercises.
• 13_key_findings_validation.sql containing checks used to reconcile the findings.